

Indian Institute of Information Technology, Vadodara

MCA Midterm Exam, September 2025

Basic of Web Programming (CS445)

Duration 2 hrs.

Max. Marks

Note: Do not write unnecessary theory; keep your answers point-to-point.

Q1. Create an HTML document that renders a simple webpage with the following output: (5 marks)

1. A main heading at the top saying "My Favorite Books".
2. Below it, an unordered list with three items: "Book 1: Web Programming", "Book 2: CSS fundamentals", and "Book 3: Java Script". The list should use bullet points.
3. After the list, a paragraph saying "These are timeless classics."
4. At the bottom, a hyperlink that says "Learn More" and links to "<https://example.com>".

Ensure the document has a proper structure with <html>, <head> (including a title "Book List"), and <body>.

Q2 Answer the following questions. (2+2 marks)

a) Suppose you have two conflicting CSS rules for a paragraph:

```
p {
  color: blue;
}
#intro {
  color: red;
}
```

The paragraph <p id="intro">Hello</p> appears in the browser. Which color will it be, and why?

b) Write the CSS rule to make all <h1> elements center-aligned.

Q3. Consider the following HTML code snippet for a simple webpage structure: (5 Marks)

```
<html>
<head>
  <title>My Page</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is a paragraph.</p>
  <ul>
    <li>Item 1</li>
    <li>Item 2</li>
  </ul>
  <a href="https://example.com">Link</a>
  
</body>
</html>
```

Identify and explain at least three errors or improvements related to HTML best practices.

For each, suggest a corrected version of the relevant code line and explain why the change is beneficial in the context of web development.

PTO

Q4. Explain the three main methods of including CSS in an HTML document. Provide a code example for each method and state one advantage and one disadvantage of each. **(6 marks)**

Q5. You are designing a navigation menu for a blog website. The menu should include links to "Home", "About", "Posts", and "Contact". Using CSS Flexbox, write a sample CSS code block to style this menu as a horizontal bar that centers the items and adds spacing between them. Then, explain how this approach differs from using CSS Grid for the same layout, including a scenario where Grid might be more appropriate. **(5 Marks)**

Q6. Imagine a webpage that displays incorrectly on mobile devices due to fixed widths. Explain how media queries in CSS can fix this, including a basic code example with a breakpoint at 600px to change a layout from horizontal to vertical. **(5 marks)**